

in the form of a high-voltage trigger pulse or a DC voltage level, to initiate the gas discharge, and a power supply with current-limiting feature to deliver the steady-state current to sustain the discharge. Amplitude of high-voltage trigger pulse needs to be greater than

the breakdown voltage of the gas mixture in question. The current-limiting feature is provided by a resistance called ballast resistance, which limits the discharge current to desired value.

There are three commonly-used circuit topologies for designing helium-neon laser power supplies. In one of the topologies, an AC-DC or DC-DC (in case of portable systems) power supply producing an output voltage slightly higher than the voltage required

to sustain the discharge plasma is cascaded to a voltage multiplier chain of diodes and capacitors to produce the starting voltage. The voltage at the output of multiplier chain drops to almost the voltage level present at the output of DC power supply due to inherently poor regulation of the multiplier chain.

In the second commonly used circuit topology, the DC power supply is connected directly to the plasma tube through the ballast resistance. A high-voltage trigger pulse of 10-15kV is applied to the tube to initiate the discharge. Once gas mixture is ionised, the DC power supply takes over to sustain the discharge.

In the third topology, constant power output fly-back converter is used. The DC to DC converter is designed to produce an open circuit output voltage of 10-15kV and a power output delivery rating equal to product of a little more than the required magnitude of DC voltage and desired discharge current. As a consequence, the converter output voltage rapidly drops to the desired value after the discharge is initiated.

Diode-capacitor multiplier chain

There are two common voltage multiplier configurations, one for the odd multiplication factors (Fig. 4), and the other for even multiplication factors (Fig. 5). The output voltage in case of multiplier configuration shown in Fig. 4 is given by:

$$V_o = \frac{nV_m}{1 + [n(n^2 - 1)/12fCR_L]}$$

In the case of multiplier chain with even multiplication factor, the output voltage is given by:

$$V_o = \frac{nV_m}{1 + [n(n^2/2 + 1)/6fCR_L]}$$

In both cases, $C_1 = C_2 = C_3 = C_4 = C_5 = C_6 = C_7 = C_8 = C$, n is the multiplication factor, f is frequency of operation equal to power-line frequency for AC-DC power supply and the switching frequency in case of a switched mode power supply. V_p is the peak ampli-

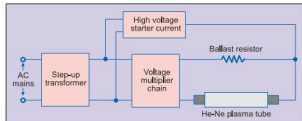


Fig. 3: Generalised block diagram of helium-neon laser power supply

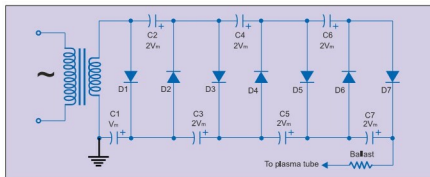


Fig. 4: High-voltage multiplier chain for odd multiplication factor

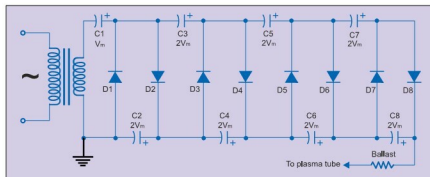


Fig. 5: High-voltage multiplier chain with even multiplication factor

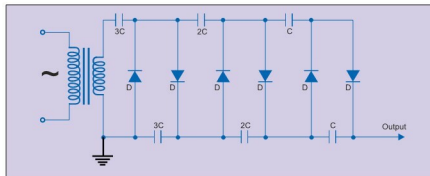


Fig. 6: Voltage multiplier chain with unequal capacitors